is $v = \begin{pmatrix} A \\ B \end{pmatrix} \in \{0,1\}^2 \xrightarrow[\text{transf}]{\text{affine}} \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ ✓

$\begin{pmatrix} (0,0) \mapsto (1,0) & (1,0) \mapsto (0,1) \\ (0,1) \mapsto (0,-1) & (1,1) \mapsto (-1,0) \end{pmatrix}$ [0x3c]

parity of x

let $B := \text{popcnt}((x \wedge 0xa\cdots a) >> 1 \wedge x) \% 2$

$A := \text{popcnt}((-x \wedge 0xa\cdots a) >> 1 \wedge -x) \% 2$

parity of -x

...

$rdx = A - B \in \{-1, 0, 1\}$

find: $rax = 1 - (A+B)$

```
mov     rbx, rax      rbx, rdx := rax_0
mov     rdx, rbx
mov     rcx, 0xaaaaaaaaaaaaaaaa   = α = Σ_{i=0}^{15} 16^i · 10   0xaa ????
and     rbx, rcx
shr     rbx, 1        rbx = (rax_0 ∧ (Σ_{i=0}^{15} 16^i · 10)) >> 1
and     rbx, rdx
popcnt  rbx, rbx
and     rbx, 1
```

$B := rbx = 1 \wedge \text{popcnt}\left(rax_0 \wedge \left(\left(rax_0 \wedge \left(\sum_{i=0}^{15} 16^i \cdot 10\right)\right) >> 1\right)\right)$

```
neg     rax           rax = -rax_0
mov     rdx, rax      rdx = -rax_0
mov     rcx, 0xaaaaaaaaaaaaaaaa   = α = Σ_{i=0}^{15} 16^i · 10
and     rax, rcx      rax = (-rax_0) ∧ α
shr     rax, 1        rax = (-rax_0 ∧ α) >> 1
and     rax, rdx      rax = ((-rax_0 ∧ α) >> 1) ∧ -rax_0
popcnt  rax, rax
and     rax, 1
```

$A := rax = 1 \wedge \text{popcnt}\left(((-rax_0 \wedge \alpha) >> 1) \wedge -rax_0\right)$

```
mov     rdx, rax      rdx :=
add     rax, rbx      rax = A + B
dec     rax           rax = (A+B) - 1
neg     rax           rax = -(A+B-1)
sub     rdx, rbx
```

// ? the point here ?

branchless 3D vector generation ?

$rdx = A - B$

$\underbrace{\{0,1\} - \{0,1\}}_{\parallel} =$ OR -1 (signed)

∞?, UNDEF/UNF

MAX OR ∞? ← $0 - 1 = -1$ is $0xff\ldots f$ = max (signed)

Bool $\begin{cases} 1-0 = 1 \text{ is } 0x00\ldots01 = 1 \\ \left.\begin{matrix} 1-1 \\ 0-0 \end{matrix}\right\} = 0 \text{ is } 0x00\ldots00 = 0 \end{cases} \in \{0,1\}$

[0x3c]

$\{-1, 0, 1\}$ ← { output IS {0, 1, MAX_INT}